



Healthcare Interoperability Meets Airflow Extensibility



Wyatt Shapiro



EHRs are the systems of record for providers



**If you build B2B healthcare software then you
need EHR integrations too**



The data transfer?

FHIR, HL7, XML, DICOM

The data standards?

ICD10, CPT, SNOMED

The endless

acronyms?!

#1 challenge in EHR integrations?

The message types?

ADT, ORM, ORU, DFT

The security protocols?

PHI, HIPAA, HITECH



~~The data transfer?~~

~~FHIR, HL7, XML, DICOM~~

~~The data standards?~~

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~~The endless~~

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#1 challenge in EHR integrations?

~~The message types?~~

~~ADT, ORM, ORU, DFT~~

~~The security protocols?~~

~~PHI, HIPAA, HITECH~~

Your customer's

time and resources!

Customers in healthcare



All the data you need lives in the EHR

But we don't have the technical expertise to get you it

But we outsource the "IT stuff" to someone else

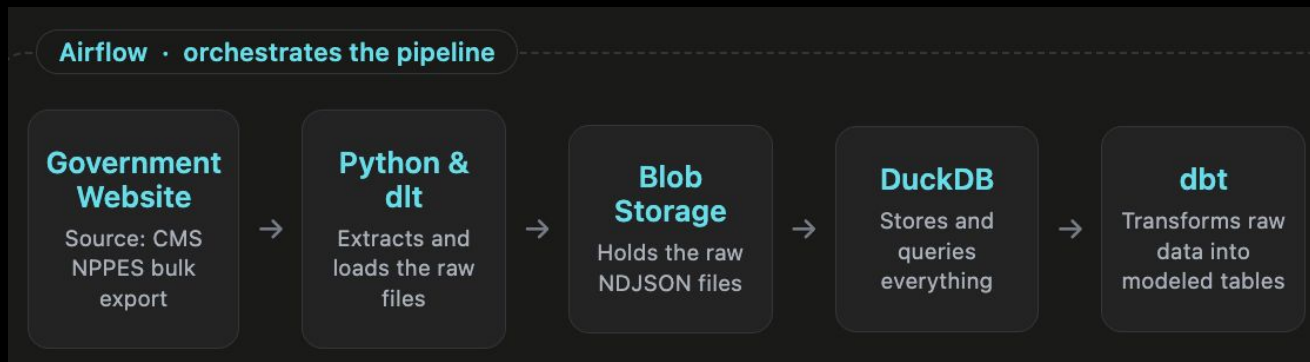
But we can get someone internally to look at this in 6+ months

But we need to get you approved by our security team

But we use multiple EHRs and systems across our facilities

...can we just email you CSV reports?

US Healthcare Clinical Workforce



Registered clinicians

7.2M

7,196,385 doctors, nurses, and other providers

Healthcare organizations

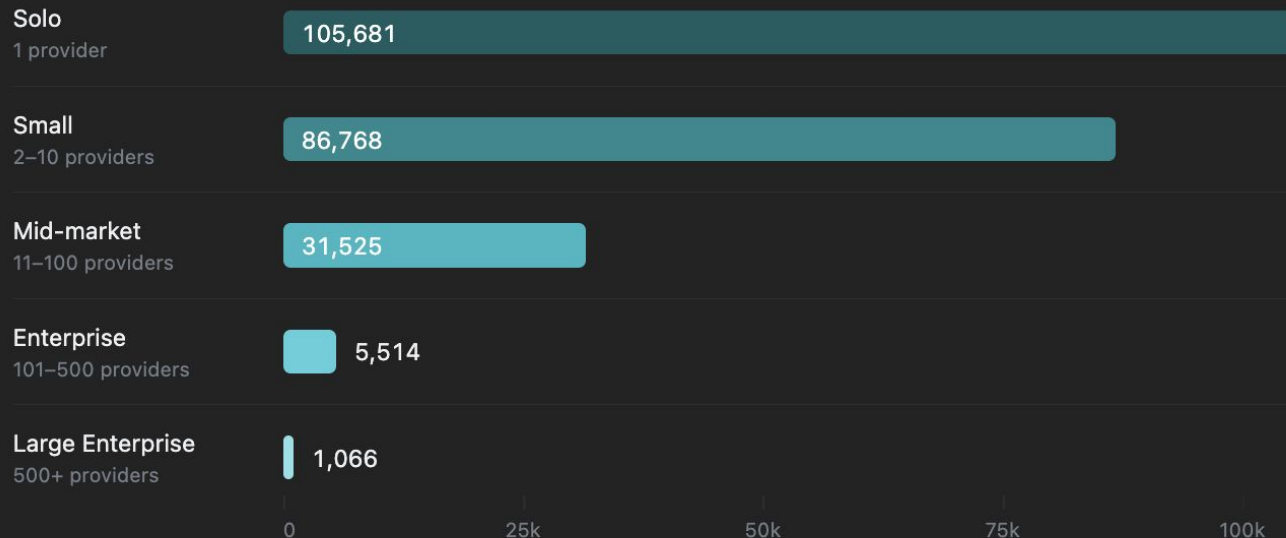
2.0M

1,999,118 practices, clinics, and facilities

Analysis and visual: Wyatt Shapiro. Data Source: CMS National Provider Directory, snapshot 2026-05-07, providers with NPIs only which may under represent institution-employed providers who bill under their org's NPI

Organizations by size tier

Covers 230,554 of the 1,999,118 healthcare organizations (12%), those where a provider can be resolved to the org via a practitioner-role record. Represents 1,931,044 of the 7,196,385 active NPIs (27%).



Analysis and visual: Wyatt Shapiro. Data Source: CMS National Provider Directory, snapshot 2026-05-07, providers with NPIs only which may under represent institution-employed providers who bill under their org's NPI



**The long tail of smaller orgs means
time & technical resources to integrate are limited
for many of your potential customers**



How do we make EHR integrations easy to design, develop, and deploy to our customers?

Start Simple: Define a Data Contract



Column	Definition	Type	Required	Notes	Standard
encounter_id	Unique identifier in the source EHR for the visit.	string	Yes	Internal ID from the source EHR.	—
provider_id	Unique identifier for the APP (NP/PA) providing care.	string or FHIR object	Yes	Internal ID, email (lowercase), or 10-digit NPI.	—
patient_id	Unique identifier for the patient.	string or FHIR object	Yes	Internal ID or MRN.	—
encounter_date	Date the medical service was rendered.	date or FHIR object	Yes	YYYY-MM-DD, local date of encounter.	ISO-8601
encounter_location	Location where the encounter occurred (patient location for virtual care).	string or FHIR object	Yes	2-character US state code (e.g., OH, GA). Required for state-specific licensing checks.	ISO-3166
encounter_status	Lifecycle stage of the encounter.	string or FHIR object	Yes	Recommend filtering for completed / finished. Values: planned, in-progress, discharged, completed, cancelled.	FHIR

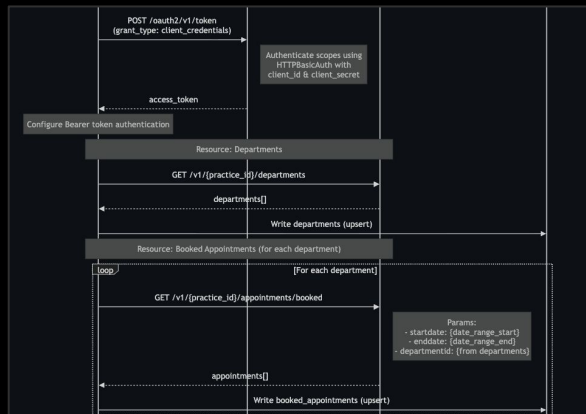
- Entities
- Column names + types
- If required or optional
- Industry standard reference
- Sample data

encounter_id	provider_id	patient_id	encounter_date	encounter_location	encounter_status	encounter_class	encounter_service_type
ENC-20260401-001	dr.olivia.foster@ziv	PAT-10234	2026-04-01	AR	completed	outpatient	primary
ENC-20260401-002	dr.olivia.foster@ziv	PAT-10587	2026-04-01	AR	completed	outpatient	primary
ENC-20260402-003	dr.william.king@zivi	PAT-10891	2026-04-02	CA	completed	outpatient	behavioral-health
ENC-20260402-004	np.kim.hawkins@zivi	PAT-10342	2026-04-02	LA	completed	outpatient	primary
ENC-20260403-005	np.kim.hawkins@zivi	PAT-10765	2026-04-03	AR	pending	telehealth	urgent-care
ENC-20260403-006	dr.william.king@zivi	PAT-10123	2026-04-03	CA	completed	inpatient	primary
ENC-20260404-007	np.james.smith@zivi	PAT-10456	2026-04-04	AR	completed	outpatient	primary
ENC-20260405-008	dr.olivia.foster@ziv	PAT-10678	2026-04-05	CA	completed	telehealth	behavioral-health
ENC-20260407-009	np.jessica.green@zivi	PAT-10901	2026-04-07	LA	completed	outpatient	primary
ENC-20260408-010	np.james.smith@zivi	PAT-10234	2026-04-08	AR	completed	outpatient	primary

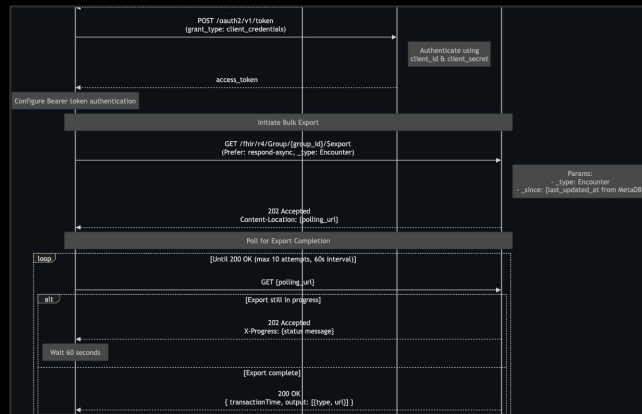
More endpoints, more problems



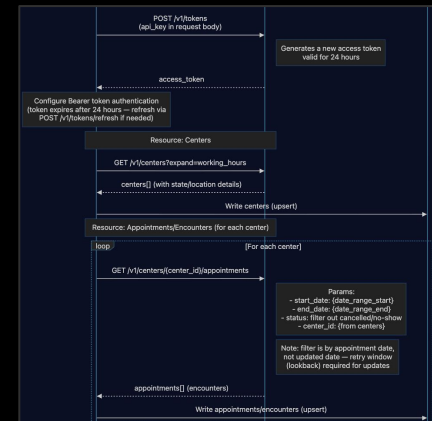
EHR 1 (Proprietary)



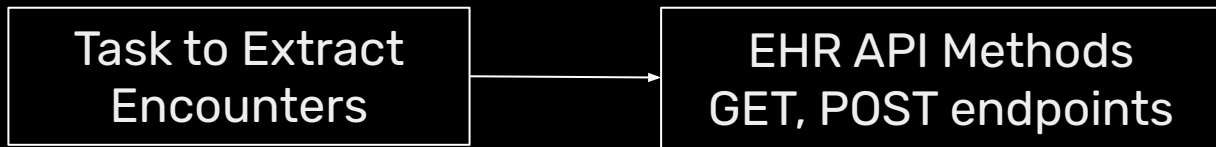
EHR 1 (FHIR Batch)



EHR 2

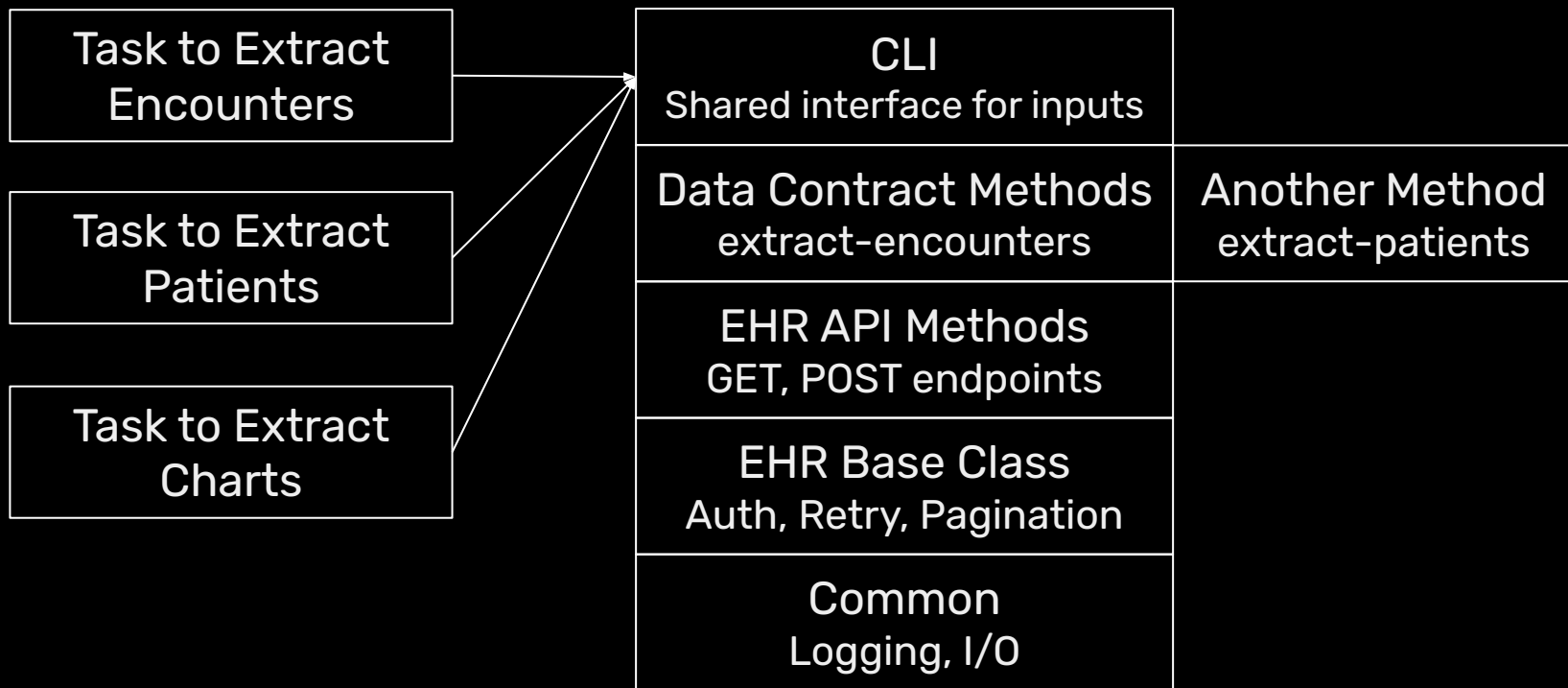


Narrowly focused

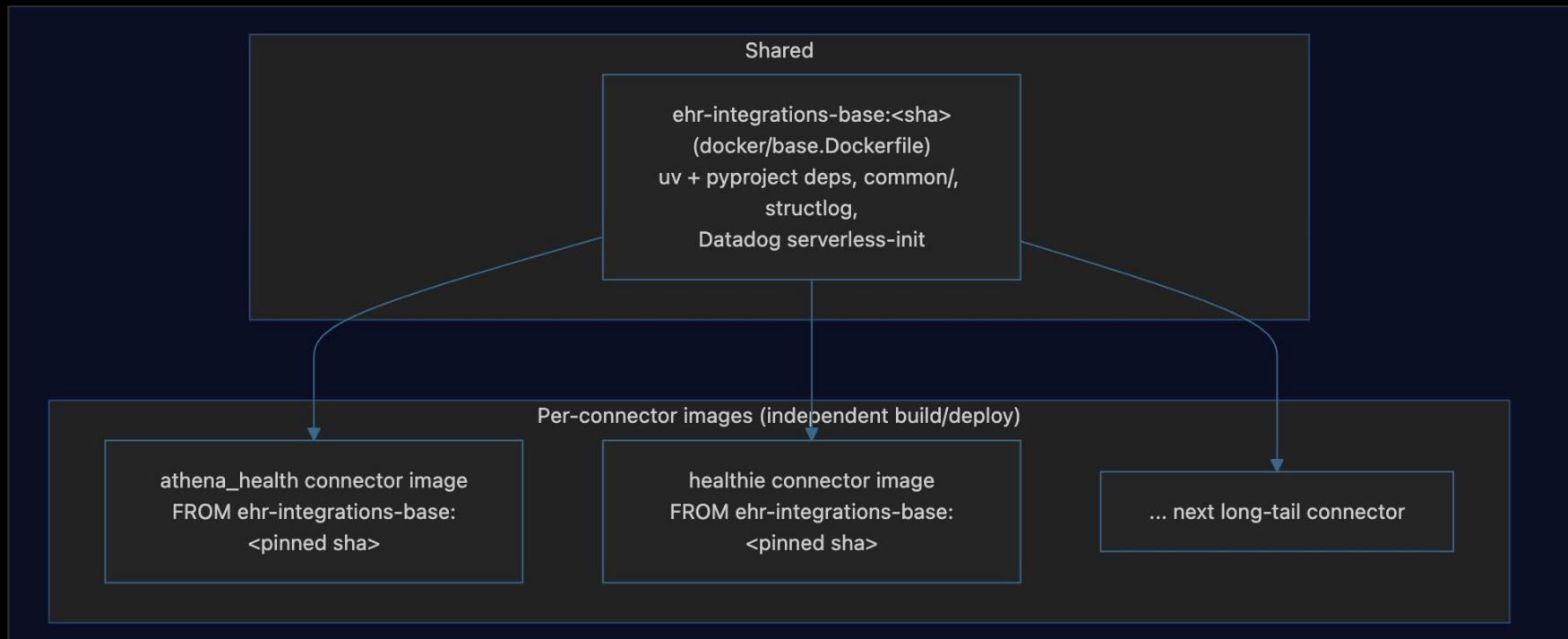


- How do we handle a new type of task for the same EHR?
- How do we handle a new customer with the same EHR?
- How do we handle a new EHR?
- How do we re-use shared code like logging?
- How do we handle non API based data extraction tasks?

Abstract API calls away



Shared Base + Independent Builds



Connector + Client Config

```
class Connector(StrEnum):
    ATHENA_HEALTH = "athena_health"
    HEALTHIE = "healthie"

CONNECTOR_IMAGES: dict[Connector, str] = {
    Connector.ATHENA_HEALTH: "ehr-connector-athena-health:<sha>",
    Connector.HEALTHIE: "ehr-connector-healthie:<sha>",
}

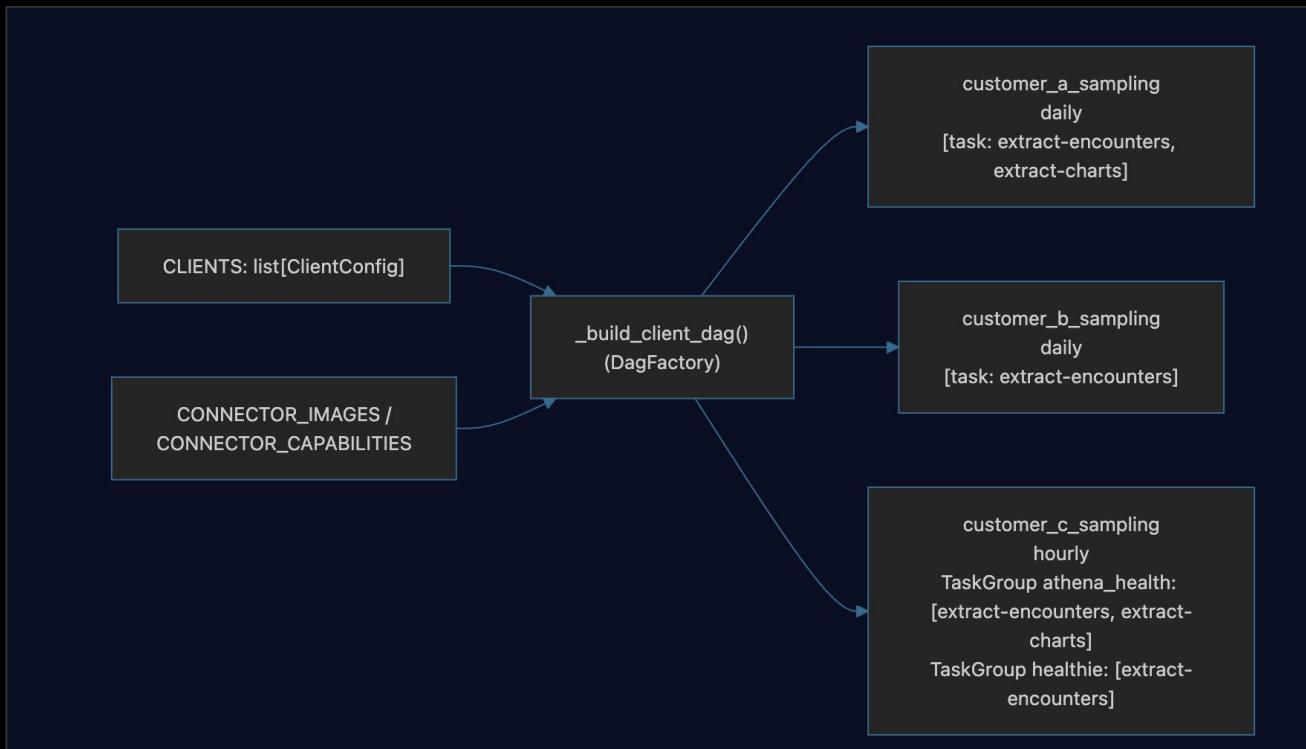
CONNECTOR_CAPABILITIES: dict[Connector, frozenset[str]] = {
    Connector.ATHENA_HEALTH: frozenset({"encounters", "charts", "patients"}),
    Connector.HEALTHIE: frozenset({"encounters", "patients"}),
}

@dataclass
class ConnectorConfig:
    connector: Connector
    entities: frozenset[str] = frozenset({"encounters"})

@dataclass
class ClientConfig:
    client_id: str
    schedule: str
    connectors: tuple[ConnectorConfig, ...]

CLIENTS: list[ClientConfig] = [
    ClientConfig("customer_a", "0 8 * * *", (
        ConnectorConfig(Connector.ATHENA_HEALTH, frozenset({"encounters", "charts"})),
    )),
    ClientConfig("customer_b", "0 6 * * *", (
        ConnectorConfig(Connector.HEALTHIE,
    )),
    ClientConfig("customer_c", "0 * * * *", (
        ConnectorConfig(Connector.ATHENA_HEALTH, frozenset({"encounters", "charts"})),
        ConnectorConfig(Connector.HEALTHIE, frozenset({"encounters"})),
    )),
]
```

DagFactory puts it all together



Running it in production



Airflow defines what is needed to be done for who, when, and in what order.

The workers (container app jobs) execute when called upon.

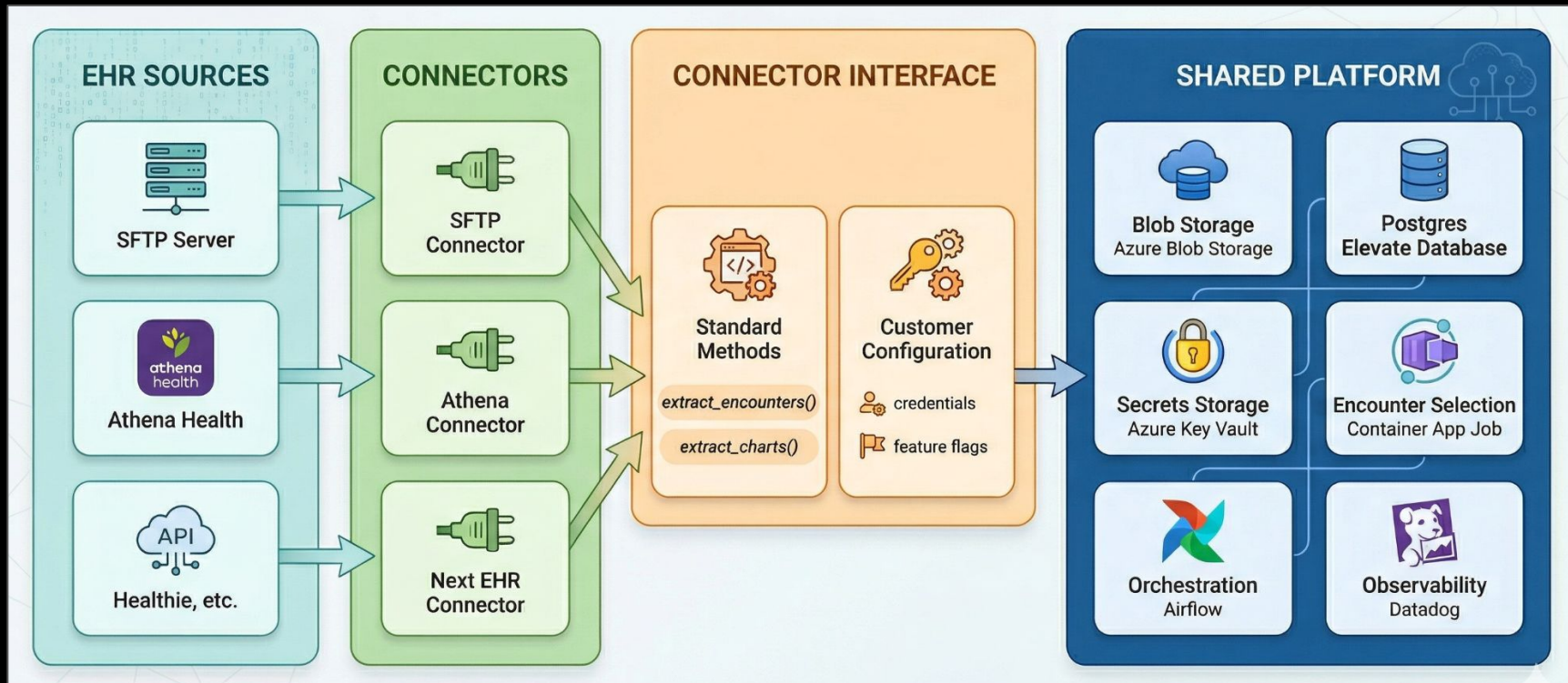
Benefits:

1. Easier code maintenance and deployment (updating one connector without affecting another connector or the orchestrator)
2. Independent tuning of compute resources based on the job type (making batch API calls vs downloading 10k PDFs)
3. Faster local development for new connectors as only Docker + Python is needed
4. Structure and re-usability without sacrificing flexibility, same CLI + contract + infra

Downsides:

1. More devops complexity (need a container registry, build/deploy setup, secrets management, logging across multiple containers)
2. Real-time is difficult given ~30 second startup times of worker containers

EHR integration platform



Indexing metadata



Open Metadata

Search for Data Assets

All ▾ Q

All Domains ▾

EN ▾

admin Default ▾

Home

Explore

Lineage

Observability ▾

Insights

Domains ▾

Govern ▾

Settings

Logout

Jump to End

Context	Total Time	Call Count	Avg Time	Min Time	Max Time
GET	18.315ms	6	3.052ms	0.681ms	7.736ms
PUT	658.726ms	3	219.575ms	23.373ms	584.572ms
Sink	2s 096.449ms	279	7.514ms	0.826ms	1s 166.740ms
Sink.PUT	1s 991.590ms	5	398.318ms	47.764ms	884.380ms
Source	4m 18s 052.413ms	288	921.616ms	5.190ms	39s 966.299ms
Source.GET	1s 183.909ms	327	3.621ms	1.495ms	145.316ms
Source.register_record	5.134ms	267	0.019ms	0.013ms	0.164ms

0 matches Search Logs

Workflow Datalake Summary:

Processed records: 279

Updated records: 0

Warnings: 0

Errors: 0

Success %: 100.0

Workflow OpenMetadata Summary:

Processed records: 281

Updated records: 0

Warnings: 0

Errors: 0

Success %: 100.0

Workflow Success %: 100.0

Workflow finished in time: 4m 22s 071.540ms

Done. Returned value was: None

Open Metadata

Search for Data Assets

All ▾ Q

All Domains ▾

Home

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Govern ▾

Settings

Logout

Data Assets

- Databases
- Dashboards
- Pipelines
- Containers
 - customstorage
 - azure_blob_test for ecrzviahealthnonprod / zivian-uat
 - Containers 3
- Governance
- Domains

Data Assets: container ▾ Domains ▾ Owners ▾ Tag ▾ Tier ▾ Certification ▾

Service Type: customstorage ▾

azure_blob_test for ecrzviahealthnonprod / zivian-uat

zivian-uat

No description

No Domains • No Owners •

azure_blob_test for ecrzviahealthnonprod / zivian-uat / charts

charts

No description

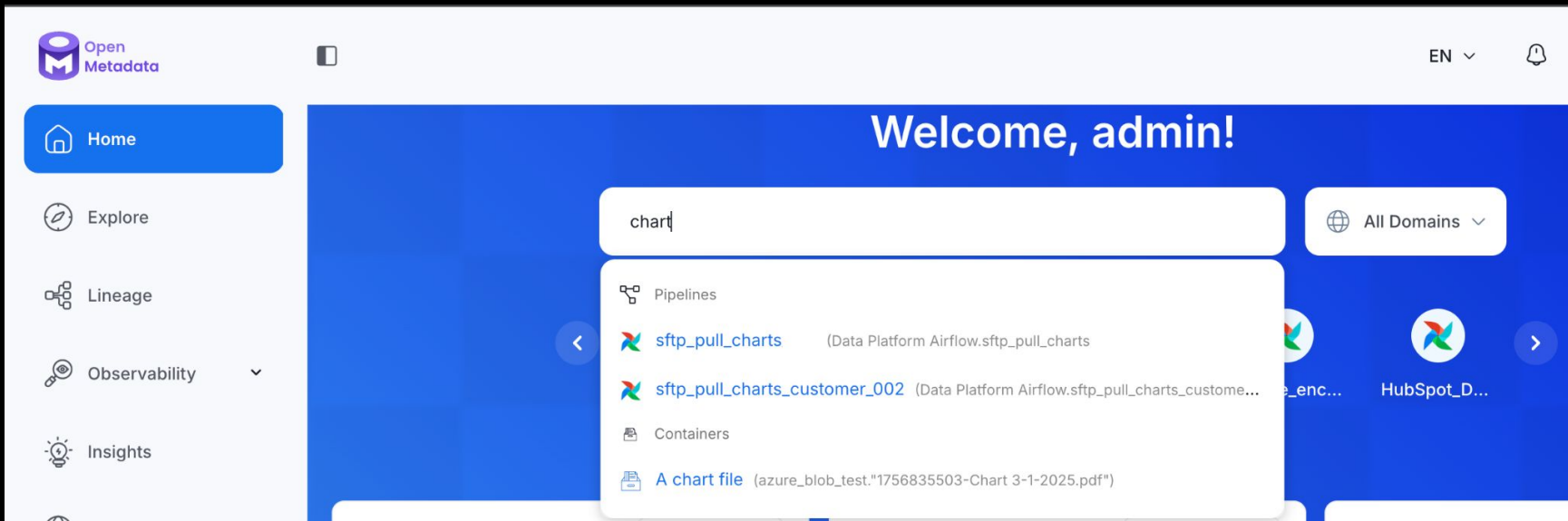
Supervising Physician • No Owners •

azure_blob_test for ecrzviahealthnonprod / charts / 1756835503 - Chart 3-1-2...

Chart 3-1-2025

Patient Chart PDF generated on 2025-03-01.

If they see it then they believe it





Questions?

Reach me at wshapiro@zivianhealth.com

Thank you to my collaborators at Zivian Health including
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